

Adapting Small Vision-Language Models (<1B) for Document Understanding

A systematic evaluation and improvement strategy

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0. Executive summary

Small VLMs (sub-1B parameters) are attractive for document understanding because documents are high-volume, privacy-sensitive, and increasingly processed on-device or at the edge — exactly where a 3B+ model is too expensive. But their document capabilities are under-characterised: most public leaderboards focus on 3B–70B models.

This report (1) selects eight sub-1B VLMs spanning generalists, edge models and OCR specialists; (2) defines a document-understanding benchmark suite anchored on **DocVQA**, **InfoVQA**, **ChartQA** and **OCRBench**, plus a **custom robustness probe**; (3) evaluates them with metrics that go beyond accuracy — **ANLS / relaxed accuracy / OCRBench score**, plus **calibration (ECE)** and **robustness retention**; and (4) turns the resulting gap analysis into a concrete, literature-grounded improvement plan.

Headline finding (from published reference figures, to be reproduced with the included pipeline): the **InternVL family at ~0.9B (InternVL2.5-1B / InternVL3-1B)** is the strongest document generalist in this size class — DocVQA $\approx$ 82–85 ANLS, OCRBench $\approx$ 785–790, ChartQA $\approx$ 75 — yet it has a **clear, consistent weak spot on InfoVQA ($\approx$ 54–56)**: multi-element, layout-and-numeric reasoning over infographics. This — not raw text recognition — is where the best small model falls short, and it is the target of the improvement strategy in Part 2.

Reproducibility note. This environment has **no GPU**, so the report does not present numbers I ran myself as if they were measured here. The **evaluation pipeline is complete, tested, and runnable** (CPU smoke-tested end-to-end; metric code unit-tested); the GPU runs are intended for a free Colab/Kaggle T4, for which exact commands are provided. Where I cite scores, they are **published reference figures** (model papers / cards) with explicit source and caveat, and the comparison table clearly separates "published reference" from "reproduced".

Part 1 — Evaluation

1. Model selection criteria

I select for **architectural diversity within the <1B budget**, so the comparison explains why one design wins, not just which wins. Three archetypes are represented:

#	Model	Params	Vision encoder + LM	Archetype	Why included
1	InternVL2.5-1B	~0.94B	InternViT-300M + Qwen2.5-0.5B, dynamic tiling	Document generalist	SoTA-for-size doc/OCR; first-class in VLMEvalKit
2	InternVL3-1B	~0.94B	InternViT-300M + Qwen2.5-0.5B, native multimodal pretrain	Document generalist	Newer pretraining recipe; head-to-head with 2.5
3	SmolVLM-500M	0.5B	SigLIP + SmolLM2, Idefics3, heavy token compression	Edge	Edge-deployment anchor; trained incl. Docmatix
4	SmolVLM-256M	0.26B	SigLIP + SmolLM2	Extreme edge	Smallest serious VLM; floor of the size/quality curve
5	LLaVA-OneVision-0.5B	~0.9B	SigLIP-SO400M + Qwen2-0.5B, AnyRes	General VLM	"General-purpose small VLM" baseline (not doc-specialised)
6	GOT-OCR2.0	~0.58B	~80M ViT + Qwen-0.5B decoder	OCR specialist	Pure transcription; tests "OCR ≠ document QA"
7	Florence-2-large	~0.77B	DaViT + BART enc-dec, task tokens	OCR/detection specialist	Unified task-token model; no free-form VQA
8	PaddleOCR-VL-0.9B	~0.9B	NaViT-style encoder + ERNIE-4.5-0.3B	Doc-parsing specialist	Newest, most document-specialised release

Rationale highlights

- **Why InternVL ×2.** It is the consensus strongest small document model, and including both 2.5 and 3 isolates the effect of the pretraining recipe at fixed architecture/size — directly relevant to "pretraining data" in the selection criteria.
- **Why two SmolVLM sizes.** They trace the size/quality frontier (256M→500M) and represent the edge-deployment use case the task emphasises; SmolVLM's training mix includes **Docmatix**, a large document instruction set, making it the strongest small-and-edge document model and a fair foil to InternVL.
- **Why a general baseline (LLaVA-OV-0.5B).** To quantify the premium document-specialised pretraining buys over a strong but general small VLM at the same ~0.9B size.
- **Why two OCR specialists (GOT, Florence-2) and a parser (PaddleOCR-VL).** Document understanding is not OCR. GOT/Florence-2 transcribe but cannot answer questions; including them makes the **"recognition vs. reasoning" gap measurable** rather than asserted, and PaddleOCR-VL tests whether a parsing-first design transfers to QA.

Architecture / pretraining / capability profiles for each model are in **Appendix A**.

2. Benchmark selection & design

Document understanding spans recognition (read the pixels), layout (where things are), and reasoning (combine fields, do arithmetic). No single dataset covers all three, so I use a **suite** plus a **custom probe**:

Benchmark	Source (VLMEvalKit / HF)	Primary metric	What it probes	Why it's here
DocVQA	lmms-lab/DocVQA (DocVQA, val)	ANLS	Printed text + layout + KIE on scanned business docs	The canonical document-QA benchmark
InfoVQA	lmms-lab/DocVQA (InfographicVQA, val)	ANLS	Dense multi-element layout, text+chart fusion, numeric reasoning	Hardest layout/reasoning; complements DocVQA
ChartQA	lmms-lab/ChartQA (test; human/aug slices)	Relaxed acc	Reading + arithmetic over plotted data	Structured-visual reasoning
OCRBench	echo840/OCRBench	OCRBench (/1000)	Regular/irregular/handwritten/artistic text, KIE, handwritten math	Fine-grained recognition capability
Robustness probe	derived from DocVQA (ours)	ANLS retention	Capture-quality + terminology stability	The "beyond accuracy" deployment axis

Why validation splits. DocVQA/InfoVQA **test** labels live behind an eval server; I use **val** so results are self-contained and reproducible. (Published model cards often quote test — a split mismatch I flag explicitly in the comparison table so numbers are not silently conflated.)

The custom robustness probe (design decisions). Real documents are phone photos, faxes, and re-compressed PDFs, and users phrase questions in domain jargon. Headline ANLS on pristine scans hides this. The probe takes N DocVQA items and emits a **paired** set — a **clean** baseline plus controlled perturbations — so failures are attributable:

- **downscale** (low-DPI / small photo → small-text legibility)
- **jpeg** quality 18 (heavy re-compression artifacts)
- **blur** (out-of-focus capture)
- **rotate** 5° (skew / handheld tilt)
- **noise** (sensor / photocopier speckle)
- **term_paraphrase** (image untouched; question rewritten with terse domain wording — "total"→"aggregate", "how much"→"what is the amount" — a deterministic, rule-based terminology stress test)

Robustness is reported as **retention = score(perturbed) / score(clean)** per family, which separates "accurate" from "accurate and stable".

3. Evaluation metrics — beyond accuracy

Metric	Definition	Why it matters for documents
ANLS	1 – normalised edit distance to best gold; 0 below 0.5	Official DocVQA/InfoVQA metric; tolerant of minor OCR/format noise (e.g. \$1,200 vs 1200) without rewarding near-misses
Relaxed accuracy	Numeric within 5% rel. error, else exact match	Official ChartQA metric; the right notion of "correct" for read-off numbers
OCRBench score	Gold string $\subseteq$ prediction, summed /1000	Standard OCRBench scoring; isolates recognition from reasoning
Calibration — ECE	Σ	$\text{acc}(\text{bin}) - \text{conf}(\text{bin})$

Metric	Definition	Why it matters for documents
Robustness retention	score(perturbed)/score(clean) per family	Production inputs are degraded; retention predicts real-world accuracy better than clean ANLS
Operational	answer-rate, load time, avg latency/sample	Edge viability: a model that's accurate but 10× slower may be unusable

Confidence for ECE is the **mean token probability** of the generated answer, read from HF `generate(output_scores=True)`. Genuinely autoregressive backends (InternVL, SmolVLM, LLaVA-OV, PaddleOCR-VL) expose it; transcription-only models (GOT, Florence-2) do not, and are reported as `ECE = n/a` rather than with a fabricated number.

4. Experimental setup & reproducibility

- **Hardware.** Free **Google Colab / Kaggle T4 (16 GB)** — every model fits in fp16/bf16; no paid hardware required, as the task allows.
- **Decoding (held constant across all models for fairness).** Greedy (`do_sample=False`, `num_beams=1`), `max_new_tokens=256`, bf16. Greedy $\Rightarrow$ deterministic $\Rightarrow$ reproducible.
- **Preprocessing.** Each model uses **its own native, documented** image pipeline (InternVL dynamic tiling up to 12×448px; SmolVLM/LLaVA AnyRes; Florence/GOT task pipelines). Forcing a shared resize would unfairly cripple tiling-based models whose document edge comes from high-res tiling — so "consistency" here means consistent decoding + consistent scoring, not identical pixels. This choice is documented so it is auditable.
- **Determinism.** Fixed seeds; greedy decoding; pinned package versions (`requirements.txt`); predictions cached to `predictions.jsonl` so re-scoring never re-runs the model and runs are resumable.
- **Fairness controls.** Same prompt template per task; same sample subset (`--limit`) across models; identical metric code for all.

5. Software stack

Tool	Role	Why
PyTorch + HF Transformers	Model loading/ inference	Every candidate ships HF weights; one API (<code>AutoModel*</code> , <code>generate</code>) covers them via thin adapters
HF Datasets	Benchmark sourcing	Canonical hosting for DocVQA/InfoVQA/ChartQA/OCRBench
VLMEvalKit	Cross-check / standard harness	The task's reference harness; we mirror its dataset choices & metrics, and provide a thin wrapper. We add a self-contained pipeline on top because VLMEvalKit reports accuracy/ANLS but not calibration or our robustness retention — the metrics that make this analysis "beyond accuracy"
PEFT (LoRA)	Part-2 fine-tuning PoC	Parameter-efficient adaptation fits the sub-1B/edge story and a T4 budget
Pillow / NumPy / torchvision	Image I/O + perturbations + tiling transforms	Lightweight, ubiquitous

Why a custom pipeline and VLMEvalKit. VLMEvalKit is the right standard for headline accuracy and is referenced for cross-checking. But calibration (ECE) and the paired robustness probe are not in its metric set, and they are central to the document deployment story — hence

the small, tested `docvlm_eval` package, whose single `evaluate.py` "loads any model, runs the benchmark, outputs per-model scores" exactly as the PoC requires.

Part 2 — Knowledge gap & improvement strategy

1. Knowledge-gap analysis (evidence-based)

Using **published reference figures** for the sub-1B models (sources & caveats in the comparison table) and the structure of the benchmark suite, three gaps emerge. The improvement strategy targets the **best model, InternVL2.5-1B**, against its own weakest axis.

Gap A — InfoVQA $\ll$ DocVQA (the primary gap). Across every model the InfoVQA score sits far below DocVQA (InternVL2.5-1B: **56.0 vs 84.8**; InternVL3-1B: 53.7 vs 81.9; LLaVA-OV-0.5B: 46.3 vs 73.7). InfoVQA is not harder to read — it is harder to reason over: dense infographics demand (i) parsing many spatially-scattered elements, (ii) fusing text with embedded charts/icons, and (iii) multi-step numeric reasoning. The 0.5B language model is the bottleneck: recognition is solved, layout-grounded reasoning is not.

Gap B — recognition $\neq$ reasoning. OCR specialists (GOT, Florence-2) are built for transcription and report **no** DocVQA/InfoVQA/OCRBench-VQA scores because they cannot answer questions; PaddleOCR-VL is evaluated only on parsing (OmniDocBench edit-distance), not VQA. This confirms that pushing OCR quality alone will not close the document-QA gap — the deficit is in reasoning over recognised content, exactly where Gap A points.

Gap C — reliability is unmeasured by leaderboards. No public card reports **calibration** or **degraded-input robustness** for these models, yet both decide deployability. Small LMs are known to be over-confident; our pipeline measures ECE and per-perturbation retention to expose it. (These are produced by running the provided pipeline; they are the columns leaderboards omit.)

Why InternVL2.5-1B is the target. It is the strongest published small document model (DocVQA 84.8, OCRBench 785, ChartQA 75.9, InfoVQA 56.0). Its DocVQA/OCRBench/ChartQA are near-saturated for the size, so the **highest-leverage, lowest-risk** improvement is its worst axis — **InfoVQA / layout-grounded numeric reasoning** (Gap A) — while hardening reliability (Gap C).

2. Improvement strategy (concrete, justified)

A staged plan, each step tied to a gap and to literature, all runnable on a single T4 — and backed by the **LoRA fine-tuning scaffold already in this repo** (`src/ocr_ft`, `scripts/finetune_lora.py`).

Step 1 — Targeted LoRA SFT on layout-reasoning data (attacks Gap A). Parameter-efficient **LoRA** (Hu et al., 2021) / **QLoRA** (Dettmers et al., 2023) on the LM + projector, freezing InternViT. Train on a reasoning-heavy document mix: **InfographicVQA train**, **DocVQA train**, **ChartQA**, **Docmatix** (the doc instruction set behind SmolVLM), and **TabMWP/PlotQA-style** numeric items. Rationale: the recognition stack is already strong; LoRA cheaply re-weights the reasoning pathway toward multi-element/numeric questions without catastrophic forgetting, and fits a T4. Keep dynamic high-res tiling on — it is what lets a 0.5B LM see small infographic text.

Step 2 — Reasoning distillation from a larger teacher (amplifies Gap A). Generate **chain-of-thought rationales** for InfoVQA/ChartQA train questions with a larger teacher (InternVL-8B/26B or a frontier VLM) and fine-tune the 1B student on (image, question → rationale → answer). Sequence-level KD / rationale distillation transfers reasoning procedure, not just answers, and is well-suited to small students (Hinton et al., 2015; Hsieh et al., "Distilling step-by-step", 2023). This directly addresses the multi-step numeric reasoning that InfoVQA exposes.

Step 3 — Robustness-aware augmentation (attacks Gap C / retention). Augment SFT with the **same perturbations as the robustness probe** (downscale/jpeg/blur/ rotate/noise) and **terminology paraphrases**. Standard augmentation theory + the paired probe give a direct read on whether retention improves. Keep augmented and clean copies so clean accuracy is preserved.

Step 4 — Calibration (attacks Gap C / ECE). Post-hoc **temperature scaling** on a held-out doc set (Guo et al., 2017) — one parameter, no accuracy change, large ECE reduction — optionally with **confidence-aware decoding** so the deployed reader can abstain/route low-confidence fields. Re-measure ECE with the pipeline.

Why this combination. LoRA+QLoRA = T4-feasible and forgetting-safe; distillation injects the reasoning the small LM lacks; augmentation + temperature scaling convert clean-set wins into deployable wins. The plan is surgical — it spends capacity on the one axis the evidence says is weak (InfoVQA/layout reasoning), not on already-saturated OCR.

3. Expected outcomes & measurement

Axis	Baseline (published ref.)	Target after plan	Measured by
InfoVQA (val ANLS)	~56	+6-10 ANLS	evaluate.py on InfoVQA
ChartQA (relaxed)	~76	+2-4	evaluate.py on ChartQA
DocVQA (val ANLS)	~85	no regression (±1)	evaluate.py on DocVQA
Robustness worst-case retention	TBD (pipeline)	+0.1-0.2	robustness probe retention
Calibration (ECE)	TBD (pipeline)	halved	ECE in summary.json

Expected magnitudes are deliberately modest and InfoVQA-focused because that is where the headroom is; the success criterion is **InfoVQA up materially with DocVQA/OCRBench held**, plus measurable reliability gains. Everything is verified by re-running the same pipeline, so before/after is apples-to-apples.

Appendix A — Model profiles

- **InternVL2.5-1B / InternVL3-1B** (OpenGVLab/InternVL2_5-1B , OpenGVLab/InternVL3-1B , MIT). InternViT-300M vision encoder + Qwen2.5-0.5B LM via an MLP pixel-shuffle projector, with **dynamic high-resolution tiling** (up to 12× 448px crops + thumbnail). Trained on large multimodal corpora with heavy OCR/document/chart data; v3 adds a native multimodal pretraining recipe. Strong OCR/doc/chart for the size.

- **SmolVLM-256M / 500M** (`HuggingFaceTB/SmolVLM-*-Instruct` , Apache-2.0). SigLIP vision encoder + SmolLM2 LM in the Llafix3 framework, with aggressive pixel-shuffle token compression for on-device use; training mix includes **Docmatix** (document instructions).
- **LLaVA-OneVision-0.5B** (`llava-hf/llava-onevision-qwen2-0.5b-ov-hf` , Apache-2.0). SigLIP-SO400M + Qwen2-0.5B with AnyRes tiling; strong general single-image/multi-image/video VLM, not document-specialised.
- **GOT-OCR2.0** (`stepfun-ai/GOT-OCR-2.0-hf` , Apache-2.0). ~80M ViT encoder + Qwen-0.5B decoder trained end-to-end for **transcription** (plain/formatted/markdown). No free-form question interface → OCR-specialist baseline.
- **Florence-2-large** (`microsoft/Florence-2-large` , MIT). DaViT encoder + BART-style enc-dec trained on FLD-5B with a **task-token** interface (`<OCR>` , `<OD>` , ...). No `<DocVQA>` token → driven with `<OCR>` ; specialist, not conversational.
- **PaddleOCR-VL-0.9B** (`PaddlePaddle/PaddleOCR-VL`). NaViT-style dynamic-resolution encoder
- ERNIE-4.5-0.3B LM, purpose-built for full-page **document parsing** (text/table/formula/chart, reading order) across 100+ languages; evaluated by its authors on parsing (OmniDocBench edit-distance), not VQA.

Appendix B — Selected references

- Mathew et al. DocVQA: A Dataset for VQA on Document Images. WACV 2021.
- Mathew et al. InfographicVQA. WACV 2022.
- Masry et al. ChartQA. ACL Findings 2022.
- Liu et al. OCRBench: On the Hidden Mystery of OCR in Large Multimodal Models. 2023.
- Chen et al. InternVL / InternVL 2.5 (arXiv:2412.05271) / InternVL3 (arXiv:2504.10479).
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- Li et al. LLaVA-OneVision. arXiv:2408.03326.
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- Xiao et al. Florence-2. arXiv:2311.06242 (CVPR 2024).
- PaddleOCR-VL Technical Report. arXiv:2510.14528.
- Hu et al. LoRA. arXiv:2106.09685. Dettmers et al. QLoRA. arXiv:2305.14314.
- Hinton et al. Distilling the Knowledge in a Neural Network. 2015. Hsieh et al. Distilling Step-by-Step. ACL 2023.
- Guo et al. On Calibration of Modern Neural Networks. ICML 2017.

Published scores quoted in this report are self-reported by model authors (papers/cards), evaluated via VLMEvalKit/OpenCompass; split conventions (val vs test) and scale conventions (OCRBench /1000 vs /100) differ between sources and are flagged in the comparison table.